Instrumental Variables (IV)

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Setup and R Packages

9.1 What is an Instrument?

Instrumental Variables (IV) represents one of the most powerful yet challenging methods in causal inference. When we cannot conduct a randomized experiment and face the dreaded problem of **unobserved confounding**, IV offers a potential path forward by exploiting "natural experiments" in observational data.

The Core Problem: Unobserved Confounding

In many real-world scenarios, we cannot assume that all confounders have been measured and controlled for. Consider these examples:

- **Education and earnings**: Unmeasured ability affects both schooling decisions and future income
- **Hospital quality and patient outcomes**: Sicker patients may systematically choose certain hospitals
- **Treatment adherence and health**: Patient motivation influences both medication compliance and health behaviors

Traditional methods like regression adjustment or matching fail when important confounders remain unobserved. This is where instrumental variables shine.

Formal Definition of an Instrumental Variable

An **instrumental variable** Z is a variable that enables causal identification of the effect of treatment A on outcome Y by satisfying three fundamental assumptions:

The Three IV Assumptions

1. **Relevance (Non-zero association)**: Z is statistically associated with treatment A

$$\text{Cov}(Z, A) \neq 0$$

2. **Exclusion Restriction (No direct effect)**: Z affects outcome Y only through its effect on treatment A

$$Z \perp Y | A, U$$

where U represents unmeasured confounders

3. **Exchangeability (No confounding of instrument)**: Z is independent of all unmeasured confounders U

$$Z \perp U$$

Intuitive Understanding: The "Nudge" Metaphor

> Think of an Instrument as a "Nudge"
>
> An ideal instrument is like a gentle nudge that pushes some people toward treatment and others away from treatment, but has no other effects on their lives. It's as if nature randomly assigned some people to receive encouragement for treatment, creating a partial experiment within observational data.

The instrument creates **exogenous variation** in treatment - variation that comes from outside the system of confounding relationships. This exogenous push allows us to isolate the causal effect of treatment from the web of confounding factors.

Classic Examples of Instrumental Variables

1. Quarter of Birth and Education (Angrist & Krueger, 1991)

- **Instrument**: Quarter of birth (Q1, Q2, Q3, Q4)
- **Treatment**: Years of education
- **Outcome**: Earnings

- **Logic**: Compulsory schooling laws mean children born in certain quarters must start school earlier/later, affecting total education, but birth quarter is otherwise random

2. Distance to Hospital and Treatment Choice

- **Instrument**: Distance to nearest specialized hospital
- **Treatment**: Receiving specialized care
- **Outcome**: Health outcomes
- **Logic**: Distance affects where you seek care but shouldn't directly affect your health (conditional on treatment received)

3. Mendelian Randomization in Genetics

- **Instrument**: Genetic variants
- **Treatment**: Biomarker levels (e.g., cholesterol)
- **Outcome**: Disease risk
- **Logic**: Genes are randomly assigned at conception and affect disease only through their effect on biomarkers

Connections to Other Causal Methods

IV differs fundamentally from other causal inference methods:

- **Unlike randomization**: We don't control the instrument assignment, we observe it
- **Unlike backdoor adjustment**: We don't need to measure all confounders
- **Unlike front-door criterion**: We don't need to identify mediating pathways
- **Like natural experiments**: We exploit quasi-random variation in nature

> ⚠ Critical Limitation Preview
>
> Unlike methods that can identify the Average Treatment Effect (ATE) for the entire population, IV typically identifies effects only for a specific subgroup - those whose treatment decisions are influenced by the instrument. We'll explore this limitation in detail in Section 9.2.

9.2 No Nonparametric Identification of the ATE

This section addresses one of the most important and often misunderstood limitations of instrumental variables: **IV cannot nonparametrically identify the population Average Treatment Effect (ATE)** in general settings.

What Do We Mean by "Nonparametric Identification"?

Nonparametric identification means we can recover a causal parameter without making strong functional form assumptions (like linearity). Methods like randomization or backdoor adjustment can nonparametrically identify the ATE - the average effect of treatment for the entire population.

The Fundamental IV Limitation

Instrumental variables face an inherent limitation: they only provide information about individuals whose treatment decisions are affected by the instrument. We learn nothing about treatment effects for people who would make the same treatment choice regardless of the instrument value.

The Information Gap

- **What IV tells us**: Treatment effects for people influenced by the instrument
- **What IV doesn't tell us**: Treatment effects for people unaffected by the instrument
- **What we want**: Treatment effects for the entire population (ATE)

Intuitive Explanation: The "Influence" Problem

Imagine studying the effect of college education on earnings using distance to college as an instrument:

- **Group A**: People who attend college regardless of distance (always-takers)
- **Group B**: People who never attend college regardless of distance (never-takers)
- **Group C**: People whose college decision depends on distance (compliers)

The instrument only creates variation for Group C. We observe what happens when Group C members are nudged toward vs. away from college, but we learn nothing about what would happen if Group A or B members changed their education status.

Why This Matters in Practice

This limitation has profound implications:

1. **Generalizability**: IV estimates may not apply to policy interventions that affect different populations
2. **External validity**: Effects may not generalize across contexts with different complier populations
3. **Policy relevance**: The ATE might be more relevant for policy than the effect for compliers

Mathematical Insight: The Identification Problem

Consider the general nonparametric case. The ATE is:

$$\text{ATE} = E[Y^1 - Y^0]$$

But IV can only identify:

$$\text{IV Estimand} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[A|Z=1] - E[A|Z=0]}$$

Without additional assumptions, this IV estimand equals a **weighted average** of treatment effects for different subgroups, where the weights depend on how responsive each subgroup is to the instrument. This generally differs from the population ATE.

Interpretation

The IV estimand:

$$\text{IV Estimand} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[A|Z=1] - E[A|Z=0]}$$

is the Wald estimator for the causal effect of a treatment (A) on an outcome (Y) using an instrumental variable (Z).

Interpretation:

- The numerator, $(E[Y|Z=1] - E[Y|Z=0])$, is the difference in average outcomes between groups defined by the instrument.
- The denominator, $(E[A|Z=1] - E[A|Z=0])$, is the difference in average treatment rates between those same groups.

Causal Meaning:

This ratio estimates the Local Average Treatment Effect (LATE): the average causal effect of (A) on (Y) for individuals whose treatment status is changed by the instrument (Z) (the "compliers"). It is valid under the standard IV assumptions: relevance, exclusion restriction, and exchangeability (plus monotonicity for LATE).

In summary:

The IV estimand isolates the causal effect of the treatment by leveraging exogenous variation in treatment induced by the instrument, thus overcoming unmeasured confounding. It tells us the effect of treatment for those whose treatment status is actually influenced by the instrument.

> When IV Can Identify ATE
>
> IV can identify the population ATE under special circumstances:
>
> 1. **Homogeneous effects**: Everyone has the same treatment effect
> 2. **Complete compliance**: Everyone's treatment is affected by the instrument
> 3. **Representative compliers**: Those affected by the instrument are representative of the population

The Trade-off: Bias vs. Assumptions

This limitation reflects a fundamental trade-off in causal inference:

- **Backdoor methods**: Can identify ATE but require "no unobserved confounders"
- **IV methods**: Can handle unobserved confounders but typically identify Local ATE

Neither assumption is necessarily better - the choice depends on which is more plausible in your specific application.

9.3 Warm-Up: Binary Linear Setting

Let's start our technical journey with the simplest possible IV setting: binary instrument, binary treatment, and linear relationships. This serves as our foundation before tackling more complex scenarios.

The Setup

Consider the following simple system:

- $Z \in \{0, 1\}$: Binary instrumental variable
- $A \in \{0, 1\}$: Binary treatment variable
- Y: Continuous outcome
- U: Unobserved confounder affecting both A and Y

The Structural Model

We can represent this system with two equations:

Treatment equation (First stage):

$$A = \alpha_0 + \alpha_1 Z + \gamma_1 U + \epsilon_A$$

Outcome equation (Second stage):

$$Y = \beta_0 + \beta_1 A + \gamma_2 U + \epsilon_Y$$

The parameter of interest is β_1 - the causal effect of treatment on outcome.

The Problem with OLS

If we simply regress Y on A, we get:

$$\hat{\beta}_1^{OLS} = \beta_1 + \gamma_2 \frac{\text{Cov}(A, U)}{\text{Var}(A)}$$

This is biased whenever treatment and the unobserved confounder are correlated ($\text{Cov}(A, U) \neq 0$).

The IV Solution: Two-Stage Least Squares (2SLS)

IV solves this by using the instrument to create "clean" variation in treatment:

Stage 1: Predict treatment using only the instrument

$$\hat{A} = \hat{\alpha}_0 + \hat{\alpha}_1 Z$$

Stage 2: Regress outcome on predicted treatment

$$Y = \beta_0 + \beta_1 \hat{A} + \epsilon$$

The IV Formula in Binary Case

In the binary setting, the IV estimator has a particularly clean form:

$$\hat{\beta}_1^{IV} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[A|Z=1] - E[A|Z=0]} = \frac{\text{Reduced Form}}{\text{First Stage}}$$

This is the **Wald estimator** - the ratio of the instrument's effect on the outcome to the instrument's effect on treatment.

Step-by-Step Numerical Example

College Education and Earnings

Let's study the effect of college completion on annual earnings using quarter of birth as an instrument.

Setting:

- Z: Born in Q4 (1) vs. other quarters (0)
- A: College degree (1) vs. no degree (0)
- Y: Annual earnings (in thousands of dollars)

Observed Data Summary (n = 1,000):

Birth Quarter	College Rate	Avg Earnings	Sample Size
Q1-Q3 (Z=0)	40%	$45k	750
Q4 (Z=1)	50%	$48k	250

Step 1: Calculate the First Stage The first stage measures how much the instrument affects treatment probability:

$$\text{First Stage} = P(A=1|Z=1) - P(A=1|Z=0) = 0.50 - 0.40 = 0.10$$

Step 2: Calculate the Reduced Form The reduced form measures the instrument's total effect on the outcome:

$$\text{Reduced Form} = E[Y|Z=1] - E[Y|Z=0] = 48 - 45 = 3$$

Step 3: Calculate the IV Estimate The IV estimate is the ratio of reduced form to first stage:

$$\hat{\beta}_1^{IV} = \frac{\text{Reduced Form}}{\text{First Stage}} = \frac{3}{0.10} = 30$$

Table 2

```
# A tibble: 2 x 4
      Z     n college_rate avg_earnings
  <int> <int>        <dbl>        <dbl>
1     0   752        0.399         44.9
2     1   248        0.492         47.4
```

Interpretation: Among individuals whose college attendance was influenced by their birth quarter, completing college increases annual earnings by $30,000 on average.

R Code Implementation

Code Explanation

- `plogis()`: Converts linear index to probability using logistic function
- We simulate the unobserved confounder `U` but wouldn't observe it in real data
- The true causal effect is 25 (coefficient on `A` in outcome equation)
- Notice how birth quarter affects both college attendance and earnings

Manual IV Calculation

```
First Stage: 0.093
```

```
Reduced Form: 2.492
```

```
Manual IV Estimate: 26.798
```

```
True Causal Effect: 25
```

```
OLS Estimate (biased): 31.189
```

```
IV Estimate: 26.798
```

Table 3: IV Regression

```
Call:
ivreg(formula = Y ~ A | Z, data = data)

Residuals:
     Min       1Q   Median       3Q      Max 
-38.0331  -8.9414  -0.1814   8.5961  45.1099 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   34.200      4.214   8.116  1.4e-15 ***
A             26.798      9.940   2.696  0.00714 ** 

Diagnostic tests:
                 df1 df2 statistic p-value  
Weak instruments   1 998     6.644  0.0101 *
Wu-Hausman         1 997     0.202  0.6530  
Sargan             0  NA        NA      NA  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.62 on 998 degrees of freedom
Multiple R-Squared: 0.5939, Adjusted R-squared: 0.5935 
Wald test: 7.268 on 1 and 998 DF,  p-value: 0.007136
```

Table 4: Comparison of OLS and IV

	Dependent variable:	
	Y	
	OLS	instrumental variable
	OLS (Biased)	IV (Causal)
	(1)	(2)
A	31.189***	26.798***
	(0.796)	(9.940)
Constant	32.347***	34.200***
	(0.517)	(4.214)
Observations	1,000	1,000
R2	0.606	0.594
Residual Std. Error (df = 998)	12.436	12.624
Note:	*p<0.1; **p<0.05; ***p<0.01	

Table 5: First Stage Regression

```
Call:
lm(formula = A ~ Z, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.4919 -0.3989 -0.3989  0.6011  0.6011 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.39894    0.01797  22.202   <2e-16 ***
Z            0.09300    0.03608   2.578   0.0101 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4927 on 998 degrees of freedom
Multiple R-squared:  0.006613,	Adjusted R-squared:  0.005617 
F-statistic: 6.644 on 1 and 998 DF,  p-value: 0.01009

First Stage F-statistic: 6.64

Strong instrument? FALSE
```

Using Built-in 2SLS

> ⚠ Common Pitfalls in Binary IV
>
> 1. **Weak instrument**: If the first stage is small, IV estimates become unreliable
> 2. **Exclusion violation**: If birth quarter directly affects earnings (e.g., through seasonal hiring), IV fails
> 3. **Defiers**: If some people do the opposite of what the instrument suggests, interpretation becomes complex

Checking Instrument Strength

9.4 Continuous Linear Setting

Moving beyond binary variables, we now consider the case where instruments and treatments can take continuous values. This setting is common in economics and health research and introduces additional nuances in estimation and interpretation.

The Continuous Linear Model

Consider the following system of equations:

First Stage (Treatment equation):

$$A = \alpha_0 + \alpha_1 Z + \gamma_1 U + \epsilon_A$$

Second Stage (Outcome equation):

$$Y = \beta_0 + \beta_1 A + \gamma_2 U + \epsilon_Y$$

Where:

- Z: Continuous instrumental variable
- A: Continuous treatment variable
- Y: Continuous outcome
- U: Unobserved confounder

IV Identification in Continuous Setting

The IV estimator in the continuous case is:

$$\hat{\beta}_1^{IV} = \frac{\text{Cov}(Y, Z)}{\text{Cov}(A, Z)}$$

This formula generalizes our binary Wald estimator. Under the three IV assumptions, this consistently estimates the causal effect β_1.

Extended Numerical Example

Distance to College and Years of Education

We'll study how years of education affect log earnings, using distance to the nearest college as an instrument.

Setting:

- Z: Distance to nearest college (0-100 miles)
- A: Years of education (8-20 years)
- Y: Log of annual earnings

Table 6: Simulation and Analysis for Continuous IV

```
  log_earnings      education        distance     
 Min.   :1.536   Min.   : 8.00   Min.   : 0.02932  
 1st Qu.:2.714   1st Qu.: 9.00   1st Qu.:26.66133  
 Median :3.061   Median :10.48   Median :52.94306  
 Mean   :3.059   Mean   :10.65   Mean   :51.71660  
 3rd Qu.:3.383   3rd Qu.:11.94   3rd Qu.:77.78682  
 Max.   :4.974   Max.   :18.20   Max.   :99.88824  
```

Reasoning:

- **Relevance**: Distance affects education costs and accessibility
- **Exclusion**: Distance shouldn't directly affect your earnings ability
- **Exchangeability**: Residential location (conditional on observables) is plausibly exogenous to ability

Simulation and Analysis

Simulation Details

- Distance reduces education by 0.03 years per mile (first stage effect)
- Each year of education increases log earnings by 0.06 (6% return)
- Ability affects both education and earnings, creating confounding
- We add realistic random variation to make the example more realistic

First Stage Analysis

Reduced Form Analysis

IV Estimation and Comparison

> ⚠ Standard Error Warning
>
> When doing manual 2SLS (Method 3 above), the standard errors from the second stage are incorrect because they don't account for the uncertainty from the first stage. Always use `ivreg()` or equivalent for proper inference.

Table 7: First Stage Analysis for Continuous IV

```
Call:
lm(formula = education ~ distance, data = df_continuous)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.9051 -1.4758 -0.2308  1.2358  8.0749 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 11.994388   0.100551  119.29   <2e-16 ***
distance    -0.025962   0.001699  -15.28   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.895 on 1498 degrees of freedom
Multiple R-squared:  0.1349,	Adjusted R-squared:  0.1343 
F-statistic: 233.6 on 1 and 1498 DF,  p-value: < 2.2e-16

First Stage Results:

Coefficient: -0.026 

Standard Error: 0.0017 

F-statistic: 233.6 

Strong instrument? TRUE
```

Table 8: Reduced Form Analysis for Continuous IV

```
Call:
lm(formula = log_earnings ~ distance, data = df_continuous)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.50614 -0.33831  0.00067  0.32951  1.81573 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  3.1875274  0.0260755 122.242  < 2e-16 ***
distance    -0.0024942  0.0004405  -5.662 1.79e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4913 on 1498 degrees of freedom
Multiple R-squared:  0.02095,	Adjusted R-squared:  0.0203 
F-statistic: 32.06 on 1 and 1498 DF,  p-value: 1.789e-08

Reduced Form Coefficient: -0.002494
```

Visualizing the IV Approach

Diagnostic Tests

> **Interpreting Continuous IV Results**
>
> In our example, the IV estimate suggests that each additional year of education increases earnings by approximately 6%. This is the causal effect for individuals whose education decisions are influenced by distance to college - typically those on the margin of attending college who face cost/access constraints.

When Continuous IV Goes Wrong

9.5 Nonparametric Identification of Local ATE

This section introduces the most sophisticated conceptual framework for understanding what instrumental variables actually identify. We'll develop new notation, explore principal stratification, and formally define the Local Average Treatment Effect (LATE).

Table 9: IV Estimation and Comparison for Continuous IV

```
Manual IV Estimate: 0.0961


Call:
ivreg(formula = log_earnings ~ education | distance, data = df_continuous)

Residuals:
      Min        1Q    Median        3Q       Max 
-1.267677 -0.293908  0.005497  0.281352  1.520081 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.03521    0.15803  12.878  < 2e-16 ***
education    0.09607    0.01480   6.491 1.15e-10 ***

Diagnostic tests:
                  df1  df2 statistic p-value    
Weak instruments    1 1498   233.596  <2e-16 ***
Wu-Hausman          1 1497     5.149  0.0234 *  
Sargan              0   NA        NA      NA    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4286 on 1498 degrees of freedom
Multiple R-Squared: 0.2551, Adjusted R-squared: 0.2546 
Wald test: 42.14 on 1 and 1498 DF,  p-value: 1.154e-10
```

Table 10: 2SLS Estimation and Comparison for Continuous IV

```
         Method Estimate
1   True Effect   0.0600
2  OLS (Biased)   0.1269
3   IV (Manual)   0.0961
4    IV (ivreg)   0.0961
5 2SLS (Manual)   0.0961
```

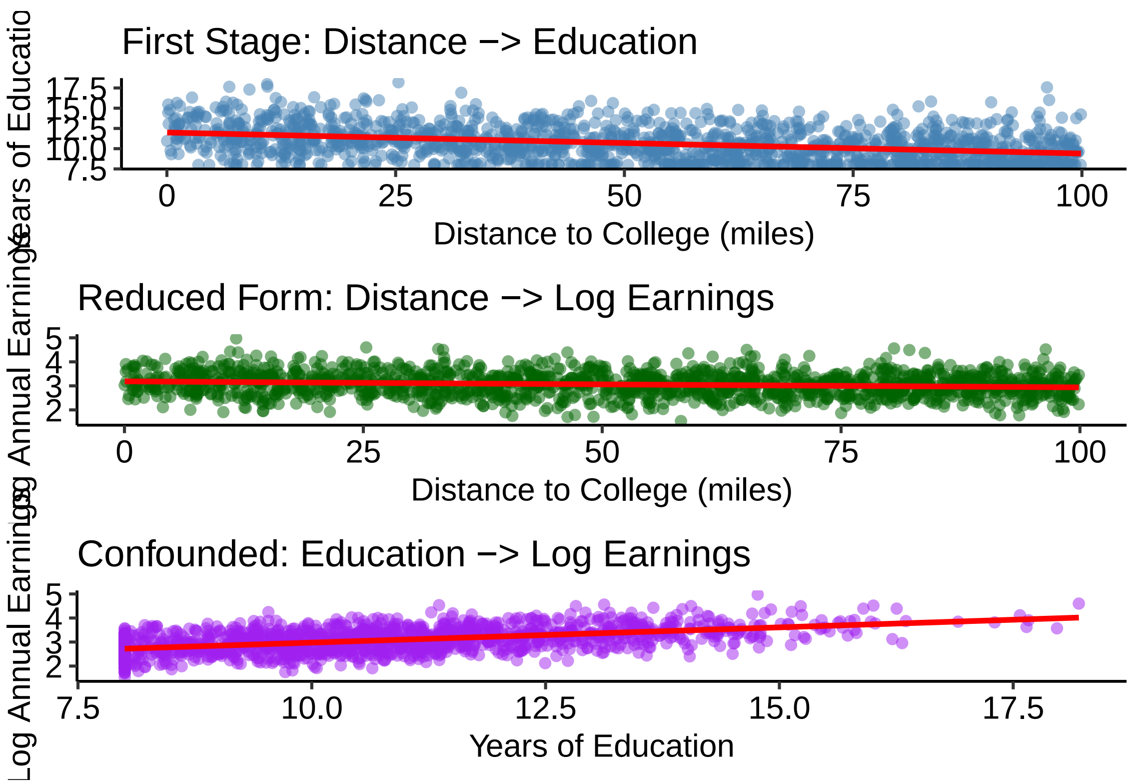

Figure 1: Visualizing the IV Approach for Continuous IV

Table 11: Diagnostic Tests for Continuous IV

```
=== INSTRUMENT STRENGTH TEST ===
First Stage F-statistic: 6.64
Critical value (rule of thumb): 10
Strong instrument? TRUE
=== INSTRUMENT EXOGENEITY CHECK ===
Reduced form coefficient: -0.002494
Implied effect (RF/FS): 0.0961
This should match our economic expectation of education returns
=== COMPARISON WITH TRUTH ===
True causal effect: 0.06
OLS estimate: 0.1269
IV estimate: 0.0961
IV bias: 0.0361
```

Table 12: When Continuous IV Goes Wrong

```
=== WEAK INSTRUMENT EXAMPLE ===
First stage F-statistic: 16.74
IV estimate: 0.0637
IV standard error: 0.0549
Notice the large standard error and unreliable estimate!
```

New Potential Notation with Instruments

To understand IV identification rigorously, we need to expand our potential outcomes framework to include potential treatments under different instrument values.

Standard Potential Outcomes: Y^a represents the potential outcome under treatment level a.

IV Potential Outcomes: We also need A^z to represent the potential treatment under instrument value z.

Expanded Notation System

For each individual i, we have:

- $Y_i^{A_i^0}$: Outcome they would have under their natural treatment when $Z = 0$
- $Y_i^{A_i^1}$: Outcome they would have under their natural treatment when $Z = 1$

- Y_i^0: Outcome under no treatment (traditional notation)
- Y_i^1: Outcome under treatment (traditional notation)
- A_i^0: Treatment they would choose when $Z = 0$
- A_i^1: Treatment they would choose when $Z = 1$

This notation recognizes that the instrument affects treatment choice, and treatment choice affects outcomes. We observe:

- $A_i = A_i^{Z_i}$ (observed treatment under observed instrument)
- $Y_i = Y_i^{A_i^{Z_i}}$ (observed outcome under observed treatment under observed instrument)

Principal Stratification

Based on their potential treatments (A_i^0, A_i^1), individuals can be classified into four **principal strata**:

The Four Principal Strata

1. **Compliers**: $A_i^1 = 1, A_i^0 = 0$
   - Take treatment when instrument is "on", don't take it when instrument is "off"
   - These are the people influenced by the instrument
2. **Always-takers**: $A_i^1 = 1, A_i^0 = 1$
   - Take treatment regardless of instrument value

- The instrument doesn't affect their behavior

3. **Never-takers**: $A_i^1 = 0, A_i^0 = 0$
   - Never take treatment regardless of instrument value
   - The instrument doesn't affect their behavior

4. **Defiers**: $A_i^1 = 0, A_i^0 = 1$
   - Do the opposite of what the instrument suggests
   - Usually assumed not to exist (monotonicity assumption)

The Monotonicity Assumption

Monotonicity (No Defiers)

We assume $A_i^1 \geq A_i^0$ for all individuals i.

This means the instrument weakly increases treatment probability for everyone. No one does the opposite of what the instrument suggests.

Monotonicity is often plausible. For example, if distance to college is our instrument, it's reasonable that distance never increases someone's probability of attending college.

Local Average Treatment Effect (LATE)

The **Local Average Treatment Effect** is the average treatment effect for the complier population:

$$\text{LATE} = E[Y_i^1 - Y_i^0 | \text{Complier}]$$

This is what instrumental variables identify under the standard IV assumptions plus monotonicity.

LATE Vs ATE: A Detailed Comparison

The **Average Treatment Effect** for the population is:

$$\text{ATE} = E[Y_i^1 - Y_i^0]$$

The **Local Average Treatment Effect** is:

$$\text{LATE} = E[Y_i^1 - Y_i^0 | \text{Complier}]$$

When LATE $\neq$ ATE

LATE differs from ATE when: 1. **Treatment effect heterogeneity**: Different people have different treatment effects 2. **Non-representative compliers**: Compliers have different treatment effects than the population average

LATE equals ATE only when treatment effects are homogeneous OR when compliers are representative of the population.

Comprehensive Numerical Example

LATE Calculation: College and Earnings Revisited

Let's work through a detailed example where we can observe the principal strata.

Population Composition: - 50% Compliers: College attendance depends on distance - 30% Always-takers: Attend college regardless of distance
- 20% Never-takers: Never attend college regardless of distance

Treatment Effects by Strata: - Compliers: $+\$35,000$ from college - Always-takers: $+\$25,000$ from college - Never-takers: $+\$45,000$ from college (hypothetical - we don't observe this)

Population ATE: $0.5 \times 35 + 0.3 \times 25 + 0.2 \times 45 = \$34,500$

LATE: $\$35,000$ (effect for compliers only)

Simulation of LATE Framework

```
=== TRUE PARAMETERS ===

Population ATE: 33.91

LATE (Compliers only): 35
```

Simulation Logic

- We create the principal strata first, then generate treatment based on strata and instrument
- Compliers attend college when close (Z=1), not when far (Z=0)
- Always-takers attend regardless of distance
- Never-takers never attend regardless of distance
- Treatment effects differ across strata to show heterogeneity

Table 13: Analyzing the LATE Framework

```
[1] "=== DATA BY STRATA AND INSTRUMENT ==="

# A tibble: 6 x 6
  strata           Z     n treatment_rate avg_outcome avg_effect
  <chr>        <int> <int>          <dbl>       <dbl>      <dbl>
1 always_taker     0   297              1        64.6         25
2 always_taker     1   311              1        65.5         25
3 complier         0   508              0        39.7         35
4 complier         1   494              1        75.1         35
5 never_taker      0   194              0        39.3         45
6 never_taker      1   196              0        41.2         45

=== IV COMPONENTS ===

First Stage (instrument -> treatment): 0.507

Reduced Form (instrument -> outcome): 18.435

IV Estimate (LATE): 36.37

True LATE: 35

IV correctly identifies LATE: FALSE
```

Analyzing the LATE Framework

Formal IV Estimation

Understanding Complier Characteristics

```
[1] "=== COMPLIER CHARACTERISTICS ==="

# A tibble: 3 x 4
  strata       proportion avg_baseline_earnings treatment_effect
  <chr>             <dbl>                 <dbl>            <dbl>
1 always_taker      0.304                  40.1               25
2 complier          0.501                  39.9               35
3 never_taker       0.195                  40.3               45
```

Table 14: Formal IV Estimation

```
Call:
ivreg(formula = Y ~ A | Z, data = df_late)

Residuals:
     Min       1Q   Median       3Q      Max 
-31.7055  -6.3188   0.2924   6.3741  27.3967 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  36.2364     0.5056   71.67   <2e-16 ***
A            36.3688     0.8336   43.63   <2e-16 ***

Diagnostic tests:
                  df1  df2 statistic p-value    
Weak instruments    1 1998     700.7  <2e-16 ***
Wu-Hausman          1 1997     107.3  <2e-16 ***
Sargan              0   NA        NA      NA    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.448 on 1998 degrees of freedom
Multiple R-Squared: 0.6974, Adjusted R-squared: 0.6972 
Wald test:  1904 on 1 and 1998 DF,  p-value: < 2.2e-16 

     Parameter Value              Interpretation
1     True ATE 33.91   Population average effect
2    True LATE 35.00   Effect for compliers only
3 OLS Estimate 29.60         Biased by selection
4  IV Estimate 36.37 Causal effect for compliers

[1] "=== RESULTS COMPARISON ==="

     Parameter Value              Interpretation
1     True ATE 33.91   Population average effect
2    True LATE 35.00   Effect for compliers only
3 OLS Estimate 29.60         Biased by selection
4  IV Estimate 36.37 Causal effect for compliers
```

Table 15: Understanding Complier Characteristics

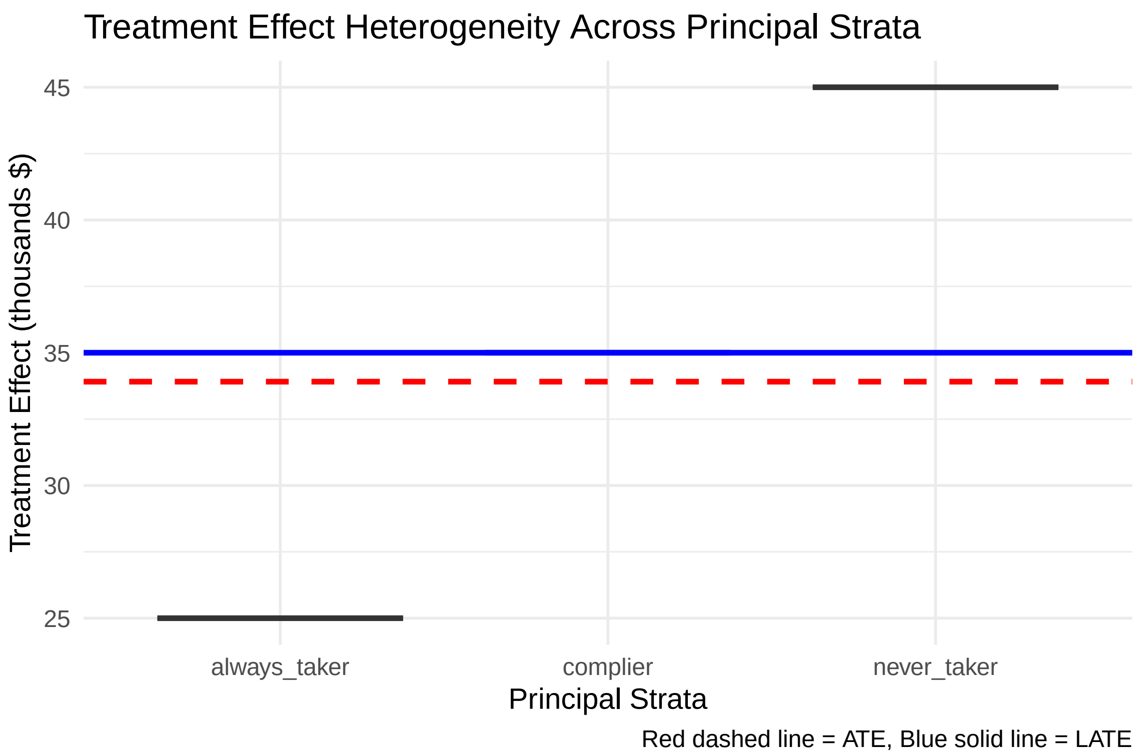

Table 16: Testing the Monotonicity Assumption

```
[1] "=== MONOTONICITY CHECK ==="

# A tibble: 3 x 4
  strata       A_when_Z0 A_when_Z1 monotonic
  <chr>            <dbl>     <dbl> <lgl>
1 always_taker         1         1 TRUE
2 complier             0         1 TRUE
3 never_taker          0         0 TRUE

All strata satisfy monotonicity: TRUE
```

Table 17: Alternative Instruments and Different LATEs

```
=== COMPARISON OF INSTRUMENTS ===

LATE with Distance Instrument: 36.3688

LATE with Aid Instrument: 0.42

Same parameter? FALSE
```

Interpretation and External Validity

LATE Interpretation Guidelines

1. **Be specific about the population**: "For individuals whose college attendance is affected by distance..."
2. **Consider generalizability**: Would policy interventions affect the same complier population?
3. **Acknowledge limitations**: "This may not represent the effect for all individuals"
4. **Context matters**: Different instruments identify different complier populations

Testing the Monotonicity Assumption

Alternative Instruments and Different LATEs

> ⚠ Different Instruments, Different LATEs
>
> Different instruments typically identify different Local Average Treatment Effects because they affect different complier populations. This is a fundamental feature of IV, not a bug - it reflects real heterogeneity in treatment effects.

9.6 More General Settings for ATE Identification

Having established the LATE framework, we now explore when and how IV methods can identify the population Average Treatment Effect (ATE) under stronger assumptions, and examine extensions to more complex settings.

Conditions for ATE Identification

While IV generally identifies LATE, the population ATE can be recovered under special circumstances:

When IV Identifies ATE

1. **Homogeneous Treatment Effects**: $Y_i^1 - Y_i^0 = \tau$ for all i
   - Everyone has the same treatment effect
   - LATE = ATE = τ
2. **Perfect Compliance**: Everyone is a complier
   - $A_i^1 = 1$ and $A_i^0 = 0$ for all i
   - No always-takers or never-takers
3. **Representative Compliers**:
   - $E[Y_i^1 - Y_i^0|\text{Complier}] = E[Y_i^1 - Y_i^0]$
   - Compliers have the same average treatment effect as the population

Parametric Approaches to ATE Recovery

When we're willing to make parametric assumptions, we can sometimes recover the ATE from LATE estimates.

Linear Response Model

Assume a linear relationship:

$$Y_i = \alpha + \beta A_i + \gamma X_i + \epsilon_i$$

Under linearity and homogeneous effects, the IV estimator directly identifies β, which equals the ATE.

Marginal Treatment Effect (MTE) Framework

The **Marginal Treatment Effect** framework allows for heterogeneous treatment effects that vary with unobserved characteristics. Under this approach:

$$\text{MTE}(u) = E[Y_i^1 - Y_i^0 | U_i = u]$$

where U_i represents unobserved heterogeneity. Different policy parameters (ATE, LATE, etc.) are weighted averages of MTE over different ranges of u.

Extensions to Nonlinear Settings

Binary Outcomes with Probit/Logit

When outcomes are binary, we might model:

First Stage: $P(A_i = 1|Z_i) = \Phi(\alpha_0 + \alpha_1 Z_i)$

Second Stage: $P(Y_i = 1|A_i) = \Phi(\beta_0 + \beta_1 A_i)$

The interpretation becomes more complex, and we typically identify local effects rather than population parameters.

Nonparametric Bounds

When parametric assumptions are questionable, we can sometimes derive **bounds** on the ATE using IV assumptions plus mild additional restrictions.

Table 18: Multiple Instruments Example: Returns to Education

```
[1] "=== MULTIPLE INSTRUMENTS COMPARISON ==="

        Instrument   Estimate         SE
1    Distance Only 0.06799591 0.06376741
2     Tuition Only 0.07881352 0.03372388
3 Both Instruments 0.07647647 0.03002470
```

Multiple Instruments

Having multiple instruments provides opportunities for:

1. **Overidentification tests**: Testing whether all instruments identify the same parameter
2. **Efficiency gains**: Combining information from multiple instruments
3. **Robustness checks**: Seeing if results are consistent across instruments

Multiple Instruments Example: Returns to Education

Consider using both distance to college and tuition costs as instruments for education:

- Both should satisfy exclusion restriction
- Both should be correlated with education (relevance)
- They may affect different complier populations

Numerical Example: Multiple Instruments

Multiple Instruments Benefits

- Using both instruments typically reduces standard errors
- Provides a robustness check - estimates should be similar if assumptions hold
- Allows for overidentification testing

Overidentification Testing

Testing for Homogeneous Vs Heterogeneous Effects

Advanced Applications

Time-Varying Instruments

Table 19: Overidentification Testing

```
=== OVERIDENTIFICATION TEST ===

Test Statistic: 0.023

Degrees of Freedom: 1

P-value: 0.8795

Reject H0 (instruments invalid)? FALSE
```

Table 20: Testing for Homogeneous Vs Heterogeneous Effects

```
=== HETEROGENEITY TEST ===

Distance Estimate: 0.068

Tuition Estimate: 0.0788

Difference: 0.0108

Z-statistic: 0.15

P-value: 0.8808

Evidence of heterogeneity? FALSE
```

In panel data settings, instruments can vary over time:

$$Y_{it} = \beta A_{it} + \alpha_i + \lambda_t + \epsilon_{it}$$

$$A_{it} = \gamma Z_{it} + \alpha_i + \lambda_t + u_{it}$$

Multiple Treatments

With multiple treatments, IV can identify effects of each treatment:

$$Y_i = \beta_1 A_{1i} + \beta_2 A_{2i} + \epsilon_i$$

This requires at least as many instruments as endogenous treatments.

Nonlinear IV with Machine Learning

Recent advances combine IV with machine learning for flexible functional forms while maintaining causal identification.

Practical Guidelines for Applied Work

Best Practices for IV Applications

1. **Justify each assumption explicitly**:
   - Provide institutional details for relevance
   - Argue for exclusion restriction with domain knowledge
   - Discuss potential confounders of the instrument
2. **Test what you can**:
   - Check first stage strength ($F > 10$)
   - Perform overidentification tests with multiple instruments
   - Examine sensitivity to functional form
3. **Interpret carefully**:
   - Acknowledge LATE vs ATE distinction
   - Describe the complier population
   - Discuss external validity limitations

4. **Report comprehensively**:
   - Show first stage results
   - Report reduced form estimates
   - Include diagnostic statistics

Common Pitfalls and How to Avoid Them

Major IV Pitfalls

1. **Weak instruments**: Use first-stage F-tests and consider weak-IV robust methods
2. **Exclusion violations**: Think carefully about all pathways from instrument to outcome
3. **Overstating generalizability**: Remember that IV identifies LATE, not necessarily ATE
4. **Ignoring heterogeneity**: Consider whether treatment effects might vary across populations
5. **Multiple testing**: With many instruments, correct for multiple comparisons

Check Your Understanding

Question 1

Explain why having multiple instruments allows for overidentification testing. What does it mean if this test rejects?

Answer

With multiple instruments, we have more moment conditions than parameters to estimate - this creates "overidentification." If all instruments are valid, they should identify the same causal parameter. The overidentification test checks whether different instruments give statistically similar estimates.

If the test rejects, it suggests that:

- At least one instrument violates the exclusion restriction
- The instruments identify different Local Average Treatment Effects (heterogeneous effects)
- There are other specification problems

Rejection doesn't tell us which instrument is problematic, but indicates the IV assumptions may be violated.

Question 2

In what sense is the LATE “local,” and why does this matter for policy evaluation?

Answer

LATE is “local” in three ways:

1. **Population**: Only applies to compliers (those affected by the instrument)
2. **Instrument-specific**: Different instruments identify different complier populations
3. **Context-dependent**: The complier population may not be representative of policy targets

This matters for policy because:

- A policy intervention might affect a different population than the instrument’s compliers
- The treatment effect for policy-relevant populations might differ from LATE
- External validity requires careful consideration of who the compliers are and whether they’re representative of the target population

For example, using distance to college as an instrument identifies effects for people on the margin due to access costs, but a tuition subsidy policy might affect a broader population.

Question 3

Under what conditions can instrumental variables identify the population ATE rather than just LATE?

Answer

IV can identify the population ATE under these conditions:

1. **Homogeneous treatment effects**: $Y_i^1 - Y_i^0 = \tau$ for all i

   - Everyone has the same treatment effect
   - LATE = ATE = τ

2. **Perfect compliance**: Everyone is a complier

   - $A_i^1 = 1$ and $A_i^0 = 0$ for all i
   - No always-takers or never-takers

3. **Representative compliers**:

- $E[Y_i^1 - Y_i^0|\text{Complier}] = E[Y_i^1 - Y_i^0]$
- Compliers have the same average treatment effect as the population

In practice, these conditions are often strong assumptions. Condition 1 rules out interesting heterogeneity, condition 2 is rare (most instruments only affect some people), and condition 3 is hard to verify. This is why most IV applications should interpret results as LATE rather than ATE unless there's strong justification for one of these assumptions.

Question 4

How would you explain to a policy maker the difference between what OLS and IV identify in the presence of unobserved confounding?

Answer

OLS with unobserved confounding:

- Compares people who chose different treatments
- Confounded by unmeasured factors that affect both treatment choice and outcomes
- Answers: "What's the difference in outcomes between those who chose treatment vs. those who didn't?"
- **Problem**: This includes selection bias, not just the causal effect

IV with valid instrument:

- Compares people who were encouraged vs. discouraged from treatment by a quasi-random factor
- Isolates the causal effect by using exogenous variation
- Answers: "What's the effect of treatment for people whose treatment decision was influenced by this quasi-random encouragement?"
- **Advantage**: Provides causal identification
- **Limitation**: Only applies to the "complier" population affected by the instrument

For policy: IV gives you the causal effect, but for a specific subgroup. You need to consider whether this subgroup is relevant for your policy intervention.

Module Summary: Key Takeaways

1. **Instrumental variables** provide causal identification when unobserved confounding is present, using three key assumptions: relevance, exclusion restriction, and exchangeability

2. **IV typically identifies LATE, not ATE**: The Local Average Treatment Effect applies only to compliers - those whose treatment is affected by the instrument
3. **Principal stratification** divides the population into compliers, always-takers, never-takers, and (assumed away) defiers based on treatment response to the instrument
4. **Different instruments identify different LATEs** because they affect different complier populations - this reflects real treatment effect heterogeneity
5. **Multiple instruments** enable overidentification testing and efficiency gains, but require all instruments to satisfy the exclusion restriction
6. **Practical application** requires careful justification of assumptions, diagnostic testing, and appropriate interpretation acknowledging the LATE framework

The power of IV lies in handling unobserved confounding, but this comes at the cost of identifying local rather than global treatment effects. Success requires strong institutional knowledge to justify the exclusion restriction and careful consideration of the policy relevance of the complier population.